

AI Ascent 2026

Sonya Huang, Pat Grady, Konstantine Buhler, Andrej Karpathy — Sequoia Capital

April 2026

Source: <https://sequoiacap.com/article/ai-ascent-2026/>

Note: This document is a structured research summary compiled from publicly available search results and analysis, archived for worldview research purposes.

Overview

Sequoia Capital's fourth annual AI Ascent conference (April 2026, San Francisco) brought together more than 150 leading founders and researchers. Speakers included Demis Hassabis (Google DeepMind), Andrej Karpathy, Greg Brockman, Boris Cherny (Anthropic), Dmitri Dolgov (Waymo), and Jim Fan (Nvidia).

The central thesis of the event: AI is a revolution in computation — not faster horses, but cars. And the cars have arrived. 2026 is the year of agents.

Pat Grady: Commercial AGI Is Here

Sequoia partner Pat Grady opened by declaring this the third inflection point in AI — functionally indistinguishable from AGI, not as a technical definition, but as a commercial and practical reality. Long-horizon AI agents — systems that can persist, recover from failure, and autonomously complete complex tasks — represent AGI in the form that matters for business.

The commercial opportunity: roughly \$10 trillion in global services revenue that the software industry has never been able to address before is now potentially in scope for AI agents. In medicine, agents can analyze genomes, recommend interventions, and identify clinical trials. In law, they negotiate contracts and settle disputes. In science, they propose novel superconductors and solve Erdős-class math problems.

Sonya Huang: The Year of Agents — Three Ingredients

Sonya Huang's thesis: if 2022–2024 was defined by chat interfaces and 2024–2025 by reasoning models, 2026 is defined by agents — a transition from systems that respond to a prompt to systems that pursue a goal.

Three ingredients have finally come together to make agents viable:

1. Models (the brain): Foundation models can now sustain productive progress on complex tasks for hours, with reasoning models dramatically extending the depth of planning and multi-step execution.

2. Tools (arms and legs): Two decades of human-built digital tools — browsers, APIs, code interpreters, file systems — have been ported directly to agents, giving them the ability to act in the world.

3. Harnesses (persistence): Orchestration frameworks, memory systems, and reinforcement learning on long-horizon tasks give agents the ability to iterate, recover from failure, and improve over time.

The combination produces agency: the practical ability to get complex things done autonomously over extended periods.

Spectrum of Agentness

Huang introduced a useful framework for understanding where organizations are on the agent adoption curve:

Tab autocomplete (2023): AI as sophisticated suggestion engine inside developer tools.

Agentic development (2024-2025): A human directs an agent or coordinates a small team of agents. The human remains the orchestrator.

Background and async agents (2026): Long-running workers that operate without continuous human supervision, often spawning sub-agents of their own. This is the current frontier.

The implication for businesses: the question is no longer whether to adopt AI but which stage of agentness to target and how to redesign workflows accordingly.

Andrej Karpathy: Software 3.0 and Agentic Engineering

Karpathy introduced a powerful three-era framework for software development:

Software 1.0: Humans write explicit code. Deterministic, inspectable, brittle.

Software 2.0: Humans train neural networks with data. The weights are the program.

Software 3.0: Humans program LLMs through context — prompts, tools, examples, memory, instructions. The context window is the program; the LLM is the interpreter.

Karpathy identified December 2025 as the personal inflection point when he went from writing roughly 80% of his own code in November to delegating 80% to agents by December — a reversal in less than one month.

He distinguished vibe coding (anyone can build software by describing what they want — raising the floor) from agentic engineering (professional-quality development using agents as force multipliers — raising the ceiling).

His most memorable insight: **'Traditional software automates what you can specify. AI automates what you can verify.'** AI capability is not evenly distributed — it spikes where there are clear data, rewards, and verification loops. This determines where AI deployment succeeds.

The AutoResearch experiment (March 2026) illustrates the pattern: Karpathy's AI coding agent ran 700 experiments in two days, discovering 20 optimizations that improved training time for a small language model. The system proposed changes, ran tests, committed improvements, and rolled back failures — autonomously. Within a month, the project had 66,000 GitHub stars.

Looking ahead, Karpathy argued that the next wave of infrastructure must be agent-native — rebuilt so the primary consumer of documentation, APIs, and deployment flows is an agent, not a human.

Implications for Organizations

The largest commercial implication of capable agents: services, not software, becomes the addressable opportunity. The shift is from selling seats to selling work.

The strongest AI startup pitch must now explain: (1) what work the company owns end-to-end, (2) why long-horizon agents make that work newly possible, and (3) why the company gets stronger as frontier models improve.

For incumbents, the question mirrors the worldview's existing framing: domain knowledge and proprietary context remain the moat — but only if converted into agent-accessible form.

Why This Belongs in the Canon

The Sequoia AI Ascent 2026 article extends several existing worldview beliefs with new precision: Belief #24 (agentic AI as the 2025-2026 inflection point) gains the three-ingredient framework as a concrete enablement model. The spectrum of agentness provides a practical adoption ladder. Karpathy's 'verify vs. specify' distinction is the most operationally useful AI capability heuristic to emerge in 2026. The December 2025 anecdote quantifies the speed of the capability shift in personal, vivid terms.